DICTO 1:2 Truncated Octahedron

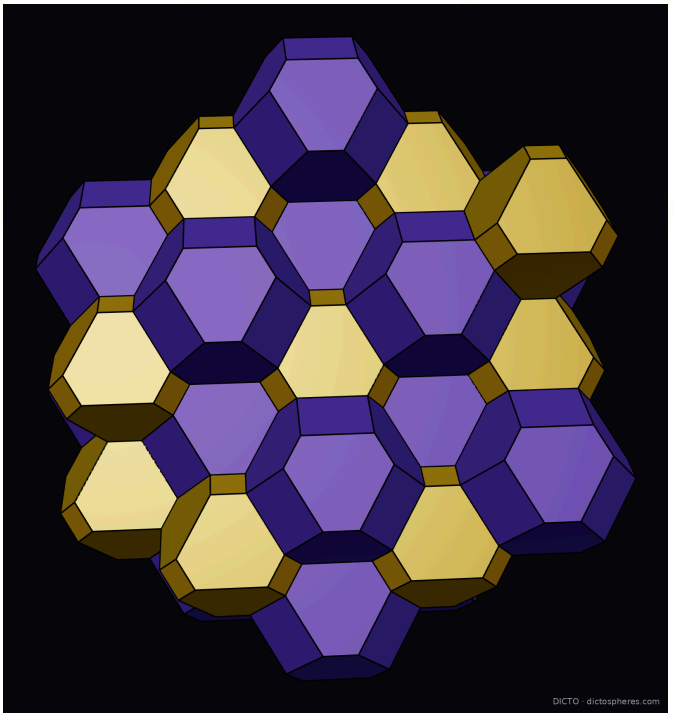
The 4D Vacuus of the Permutohedron



the web of all orderings of five things, its rules made solid: 65 central slices in four shapes, five truncated octahedra at the heart slotted along regular hexagons, a count for every dimension, and DICTONIC's golden bond as a walk through permutohedra

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Paper #105 · Version 1 · 11 November 2026 · DICTO DOI: 10.5281/zenodo.24121177 (all versions; Version 1: 10.5281/zenodo.24121178) · Licence CC BY-SA 4.0 · Data, scripts and checks: https://github.com/DICTOR-Master/kaleidohedra/tree/242cf18c/studies (folder 2026-11-10-permutohedron) · referee reports and judgements: papers/2026-11-11-4d-vacuus
DICTO: ideas, names, constructions and direction.
Computation, verification and drafting with Claude (Anthropic).

Version 1. Theorems 1–4 are proved here, each count also checked by exact computer arithmetic (rational and $\mathbb{Q}(\sqrt{5})$); checked before release by an in-house panel of four automated referees and one independent attack (reports and judgements kept with the paper's data).



DICTO 1:2 Truncated Octahedra (gold) fill space with their partners, DICTO 2:1 Truncated Octahedra (violet; squares of side 2, hexagons with sides 2 and 1 in turn): the stop x = 1/2 of DICTONIC's golden bond (Section 5). Unlike the 1:1, the 1:2 does not fill space by shifting alone; its partner is the 2:1.

Write the numbers 1 to 5 in every possible order: 12345, 12354, …, 54321. There are 120 orders. Link two of them whenever one becomes the other by swapping two neighbouring values, 2 and 3 say. The result is a web of 120 knots and 240 exchanges, and it is the skeleton of a four-dimensional solid, the 4D permutohedron. Its smallest rings are its faces: squares, where two swaps do not disturb each other, and hexagons, where three swaps go round and come back (Figure 1). This paper builds the 4D Vacuus of that solid: the few flat slices through its centre that between them carry every face, and so every rule of the web.

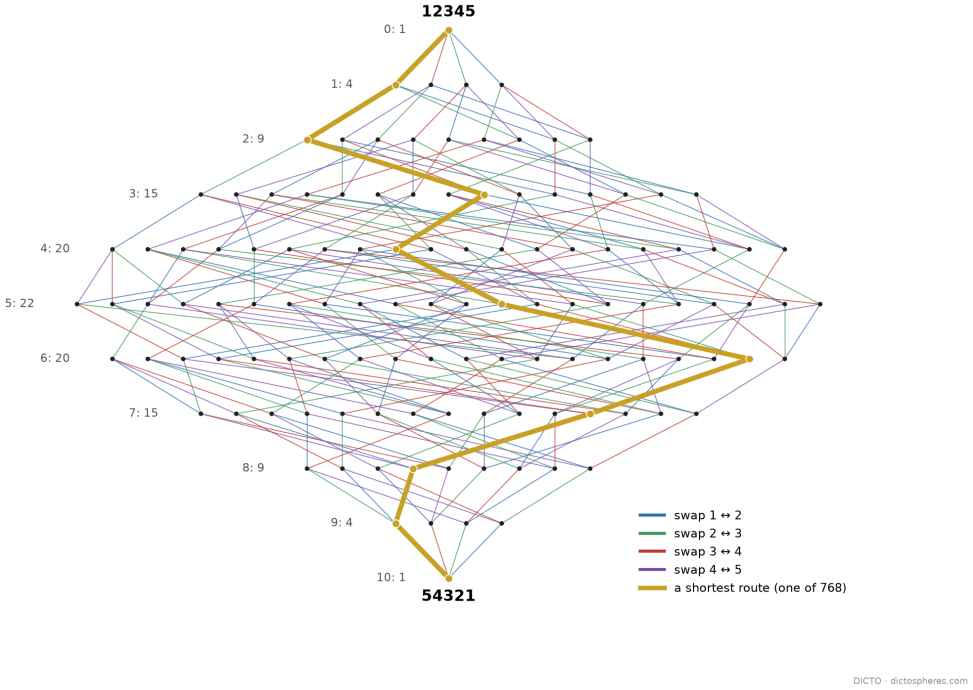


Figure 1. The web of exchange. Rows: the number of pairs out of order (1, 4, 9, 15, 20, 22, 20, 15, 9, 4, 1 orderings). Bond colours: which neighbouring values are swapped. Gold: one of the 768 shortest routes from 12345 to 54321.

1. The web of exchange

The 4D permutohedron is the convex hull of the 120 points $(p_1, \ldots, p_5)$ whose coordinates are 1, …, 5 in some order; it lies in the 4-space $p_1 + \ldots + p_5 = 15$, centred at (3, 3, 3, 3, 3). It is simple (four edges at each corner) and has 240 edges, 150 two-dimensional faces (60 hexagons, 90 squares) and 30 cells (10 truncated octahedra, 20 hexagonal prisms). Its edges join orders that differ by swapping two neighbouring values, so its skeleton is the web above: four exchanges at every knot, in ten directions. The following facts are classical and were recounted exactly (web_of_bonds.py): the number of exchanges between two orders is the number of pairs they put in opposite order; 54321 is farthest from 12345, ten exchanges away, along 768 shortest routes (Stanley's count of reduced words of the longest permutation); the 2-faces are the web's rings, a square for two swaps of disjoint value pairs and a hexagon for the braid ring $s_k s_{k+1} s_k = s_{k+1} s_k s_{k+1}$. In #104 the same solid was swept by a moving 3-space: 26 cells are born, 66 pairs swap and 26 cells die, and these are the numbers of orders with one, two and three descents (1, 26, 66, 26, 1).

2. The 4D Vacuus

The *Vacuus* of a centrally symmetric polytope is the set of hyperplanes through its centre spanned by its ridges (the faces one dimension below the cells), each ridge lying in exactly

one; a *slice* is the polytope cut by one of them. DICTO's Vacuus solids are built from such central slices: the octahedron from three squares through its centre that carry all its edges, the cuboctahedron from four hexagons. One dimension up the slices are 3D and carry the 2-faces. A 2-face of the 4D permutohedron does not pass through the centre, so together with the centre it spans exactly one 3-space through the centre: its *central slice*. The 4D Vacuus is the set of these slices.

A 2-face is an ordered split of the positions {1, …, 5} into three blocks B_1, B_2, B_3 of sizes (a, b, c): the positions in B_1 hold the a smallest values in any order, those in B_2 the next b, those in B_3 the largest c. Its central slice has the normal vector that takes the value $c_1 = bc(b + c)$ on B_1, $c_2 = -ac(a + 2b + c)$ on B_2 and $c_3 = ab(a + b)$ on B_3 (it is orthogonal to (1, …, 1), constant on each block, and orthogonal to the vector of centred block sums $S_1 = -a(b + c)/2$, $S_2 = b(a - c)/2$, $S_3 = c(a + b)/2$, so its hyperplane passes through the face; checked against exact elimination for every 2-face).

Theorem 1 (the 4D Vacuus). The 150 two-dimensional faces of the 4D permutohedron lie in exactly 65 central slices, each face in one. Up to the symmetries of the solid (the 120 permutations of the coordinates and the central inversion) there are four slices:

slices	split (a, b, c)	normal (values on B_1 | B_2 | B_3)	slice shape	edge lengths²	faces of the 4D solid carried
30	(1, 2, 2)	(8 | −7 −7 | 3 3)	32 corners, 52 edges; 14 four-sided faces (2 of them squares) + 8 hexagons	eight values, 13/18 to 8	2 squares
20	(1, 1, 3)	(6 | −9 | 1 1 1)	truncated-octahedron type: 6 rectangles + 8 hexagons (2 regular)	2 and 56/9	2 hexagons
10	(1, 3, 1)	(3 | −2 −2 −2 | 3)	24 corners, 42 edges; 18 four-sided faces (6 squares, 12 trapezoids) + 2 regular hexagons	2, 4, 8	2 hexagons
5	(2, 1, 2)	(1 1 | −4 | 1 1)	the DICTO 1:2 Truncated Octahedron (permutohedron of 1, 2, 4, 5): 6 squares + 8 hexagons	2 and 8	6 squares

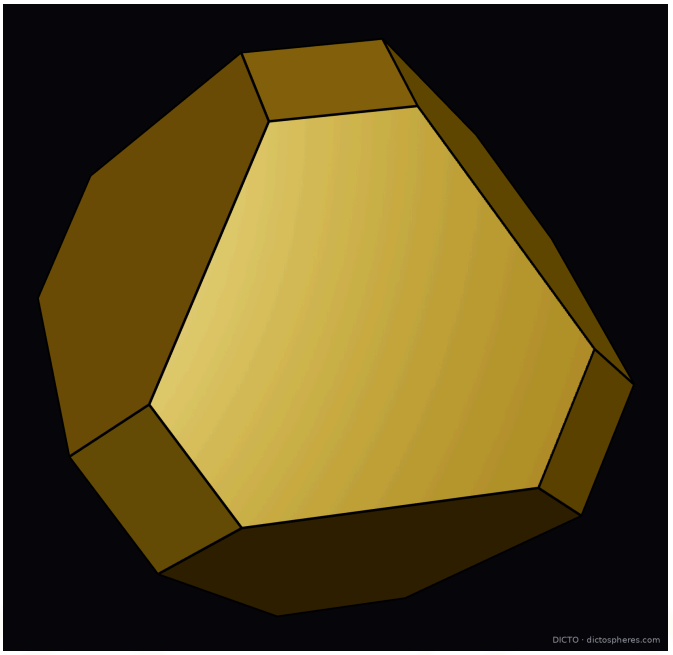
Proof. The normal formula shows that a slice depends only on the blocks and the values (c_1, c_2, c_3); since $c_1 > 0 > c_2$, for five values two splits give the same slice only if they are reverses of each other (B_3, B_2, B_1), or, when a = c, have the same middle block (for an even number of values there is one more coincidence, Theorem 3). Counting: splits (1, 1, 3) and (1, 2, 2) give 5!/(1!1!3!) = 20 and 5!/(1!2!2!) = 30 slices; (1, 3, 1) and (2, 1, 2) give C(5, 3) = 10 and C(5, 1) = 5; in all 65, and 30·2 + 5·6 = 90 squares, 20·2 + 10·2 = 60 hexagons. Slices with the same split are images of one another under a permutation of the coordinates, so they are congruent. The shapes were computed in exact rational arithmetic: corners are the solid's corners on the slice and the points where edges cross it; every face proposed by a hull routine was confirmed exactly (all its corners on one plane, every other corner strictly on one side), and every shape has Euler number 2. ∎

An earlier floating-point count split the 30 congruent slices of the first row into several apparent types; the exact computation above supersedes it.

3. The heart: five truncated octahedra

The five slices of the last row are the slices $p_i = 3$: the middle value held in one place while the other four take 1, 2, 4, 5 in every order. Each is the 3D permutohedron of the values 1, 2, 4, 5: the **DICTO 1:2 Truncated Octahedron**, a truncated octahedron in shape (the same faces meeting in the same way as Archimedes', but not his equal-edged solid) with squares of

side 1 and hexagons whose sides are 1 and 2 in turn (√2 and 2√2 in these coordinates), and it carries the six squares of the 4D solid in which the middle value stays put. These five are the heart of the Vacuus (Figure 2).



The DICTO 1:2 Truncated Octahedron close up: each hexagon has sides 1, 2, 1, 2, 1, 2; the squares have side 1.

Theorem 2 (how the heart is slotted together). Any two of the five central truncated octahedra, $p_i = 3$ and $p_j = 3$, cross in a regular hexagon through the centre, with side equal to its radius 2√2; its corners are the points with $p_i = p_j = 3$ and the other three coordinates 1, 3, 5 in any order. Three of them share a segment through the centre, and four or five share only the centre.

Proof. A point lies in the permutohedron exactly when its coordinates sum to 15 and its k largest coordinates sum to at most 5 + 4 + … (the k largest values), for every k (majorization; Rado 1952, after Hardy, Littlewood and Pólya). With $p_i = p_j = 3$ and the other three q_1, q_2, q_3 summing to 9, these conditions reduce to $1 \le q_m \le 5$: a hexagon with corners the orders of (1, 3, 5), each at distance √8 from the centre and from its neighbours. The plane meets no corner or edge of the solid; its six corners are where it pierces 2-faces, which was confirmed exactly by intersecting the plane with all 150 faces. Three conditions $p_i = p_j = p_k = 3$ leave $q_1 + q_2 = 6$ with $1 \le q \le 5$, a segment; four leave the centre. ∎

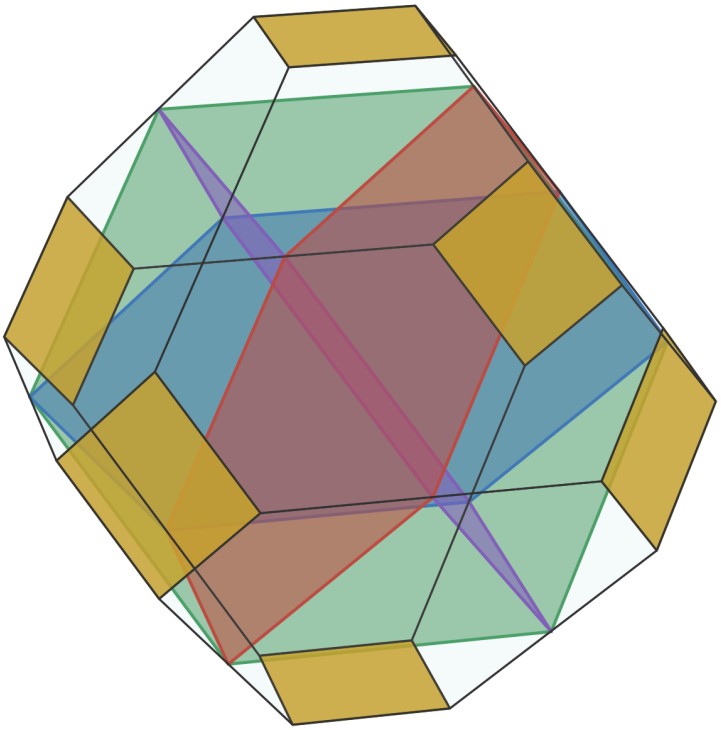


Figure 2. One of the five central slices as a see-through shell: its six squares (gold) are faces of the 4D solid; the four coloured regular hexagons through its centre are where the other four central slices cross it. The cardboard octahedron Vacuus slots three squares along their diagonals; the 4D heart slots five truncated octahedra along ten regular hexagons.

The heart nests the dimensions inside one another: the 4D permutohedron of 1, …, 5 holds at its centre, five times, the 3D permutohedron of the four values that remain when the middle one is taken out; in the same way the hexagon of the orders of 1, 2, 3 holds the segment between the orders of 1 and 3. This works whenever the number of values is odd, so that there is a middle value to hold still. (For any value v the hyperplane $p_i = v$ meets no edge inside it, since an edge changes two coordinates from k, k + 1 to k + 1, k through non-integers; so the slice $p_i = v$ is the permutohedron of the other n − 1 values, and it passes through the centre exactly when v is the middle value.)

4. The Vacuus in every dimension

The construction works for the permutohedron of 1, …, n in dimension n − 1: its ridges (faces one dimension below the cells) are the ordered splits into three blocks, and each spans one central slice with the centre.

Theorem 3 (the Vacuus count). The ridges of the permutohedron of 1, …, n lie in exactly N(n) central hyperplanes, where
$N(n) = \Sigma_{a<c}\, n!/(a!\, b!\, c!) + T(n)$, the sum over a, b, c ≥ 1 with a + b + c = n, and $T(n) = 2^{n-1} - 1$ for n odd, $2^{n-2} - 1$ for n even.

n (dimension)	3 (2)	4 (3)	5 (4)	6 (5)	7 (6)	8 (7)	9 (8)
ridges	6	36	150	540	1806	5796	18150
Vacuus slices N(n)	3	15	65	225	770	2443	7761

Proof. By the normal formula, a split with a ≠ c gives a slice whose normal takes three different values; the normal is unique up to scale, its level sets are the blocks, its one negative level is B_2, and the two positive levels are B_1 and B_3 in either order, so only the split and its reverse give the slice, and no two-valued slice can equal it: the first sum. A split with a = c gives a normal with two values, c_1 on $S = B_1 \cup B_3$ (of even size 2a) and c_2 on the middle block T; the ratio $c_1 : c_2 = -b : 2a$ depends only on the sizes of S and T, so the slice is fixed by the unordered pair {S, T}, and every pair with |S| even and at least 2 occurs. When n is odd exactly one of S, T has even size: $2^{n-1} - 1$ pairs. When n is even, |S| = 2a is even so |T| is even too, and the split (a, b, a) gives the same slice as (b/2, 2a, b/2) with S and T exchanged: the pairs with both parts even, $(2^{n-1} - 2)/2 = 2^{n-2} - 1$. The two-valued slices are the hyperplanes $\Sigma_{i\in S}\, x_i = 0$ through the centre (centred coordinates), parallel to facets. In closed form, with c_n the central trinomial coefficient (1, 1, 3, 7, 19, 51, …; OEIS A002426), the splits with a = c number $A_n = c_n - 1 - [n \text{ even}]\cdot C(n, n/2)$, and $N(n) = (3^n - 3\cdot 2^n + 3 - A_n)/2 + T(n)$. Brute force over every ridge for n = 3, …, 9, with each slice found by exact elimination, gives the same numbers (vacuus_general.py). ∎

In two dimensions the Vacuus of the hexagon is its three long diagonals; in three, the truncated octahedron's 36 edges lie on 15 planes through its centre. Neither N(n) nor its three-valued part (0, 12, 50, 210, 707, 2380, 7506) is in the On-Line Encyclopedia of Integer Sequences (searched 11 November 2026 for 3, 15, 65, 225, 770, 2443 and for 12, 50, 210, 707, 2380); we make no stronger claim of novelty.

5. DICTONIC: the golden bond walks through permutohedra

DICTONIC's golden bond (#48, Theorem 4) uses the family of semi-uniform truncated octahedra T(x), the hull of all signed orders of (1 + x, x, 0) (Koca et al. 2015: the orbit of the octahedral group with Dynkin labels (1, x, 0)), which with T(1/x) on the cube centres fills space. It is a walk through permutohedra. At x = 1/2 the partner T(2) is the **DICTO 2:1 Truncated Octahedron**: squares of side 2, hexagons with sides 2 and 1 in turn, the permutohedron of −5, −1, 1, 5; the two solids' hexagons are congruent, and along a shared hexagon each edge of the 1:2 between two hexagons meets an edge of the 2:1 between a hexagon and a square. The 1:2 alone cannot fill space by translations: its hexagons are not centrally symmetric (side 1 is opposite side 2), and a convex solid that fills space by translations has centrally symmetric facets (Minkowski for lattices; Venkov, McMullen in general). Whether turned and mirrored copies can do it is not settled here.

Three solids. Read by the ratio of square side to hexagon side, the slider holds three simplest members: the 1:1 (Kelvin's truncated octahedron, x = 1), which fills space alone and is its own partner; and the DICTO 1:2 and DICTO 2:1 Truncated Octahedra (x = 1/2 and x = 2), mirror images of each other in proportion, which fill space only together. (A 2:2 is the 1:1

at twice the size: only the ratio decides the shape.) Every other ratio on the slider pairs with its reverse in the same way.

Theorem 4. For every $x > 0$, $T(x)$ is, up to scale, the permutohedron of four values placed symmetrically about zero, $-q, -p, p, q$, with $x = (q - p)/(2p)$. In particular: $x = 1$ (Kelvin's truncated octahedron) is the permutohedron of $-3, -1, 1, 3$, equally spaced, the 3D permutohedron itself; $x = 1/2$ is the permutohedron of $-2, -1, 1, 2$, which is $1, \ldots, 5$ without its middle value: the DICTO 1:2 Truncated Octahedron, the heart of the 4D Vacuus (Theorem 2); and $x = \tau$ (the golden bond) is the permutohedron of $-\tau^3, -1, 1, \tau^3$. Equivalently, $T(x)$ is the orbit of the weight with A_3 Dynkin labels $(x, 1, x)$, whose full octahedral symmetry is the diagram symmetry of equal outer labels.

Proof. The map $y \mapsto \tfrac{1}{2}(y_1 + y_2 - y_3 - y_4, y_1 - y_2 + y_3 - y_4, y_1 - y_2 - y_3 + y_4)$ is an isometry of the 3-space $y_1 + \ldots + y_4 = 0$ onto ordinary space. It sends the order $(q, p, -p, -q)$ to $(p + q, q - p, 0)$, and the 24 orders of $-q, -p, p, q$ to the 24 signed orders of $(p + q, q - p, 0)$, which are the corners of $T(x)$ scaled by $(q - p)/x$ when $(1 + x)/x = (p + q)/(q - p)$, that is $x = (q - p)/(2p)$. For $x = \tau$, $q = p(1 + 2\tau) = p\tau^3$. Checked exactly for $x = 1/2$, 1 and τ by comparing all 276 distances between corners (dictonic_family.py). ∎

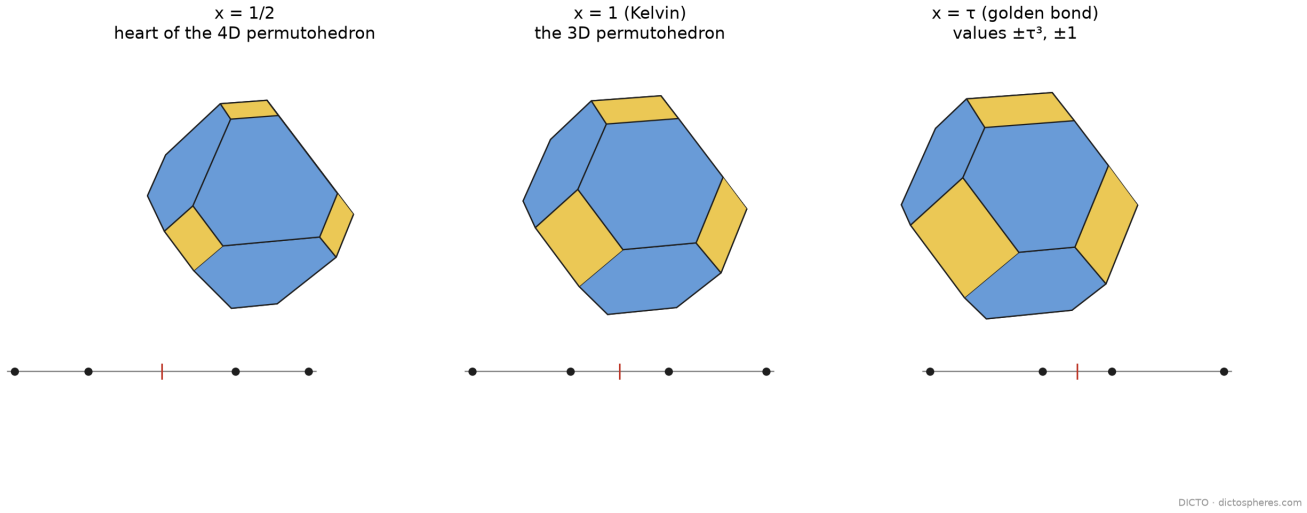


Figure 3. Three stops of DICTONIC's golden slider, each the permutohedron of the four values on the line beneath it (the red mark is the centre). At $x = 1/2$ the values are $1, \ldots, 5$ without the middle: the slider passes through the heart of the 4D permutohedron.

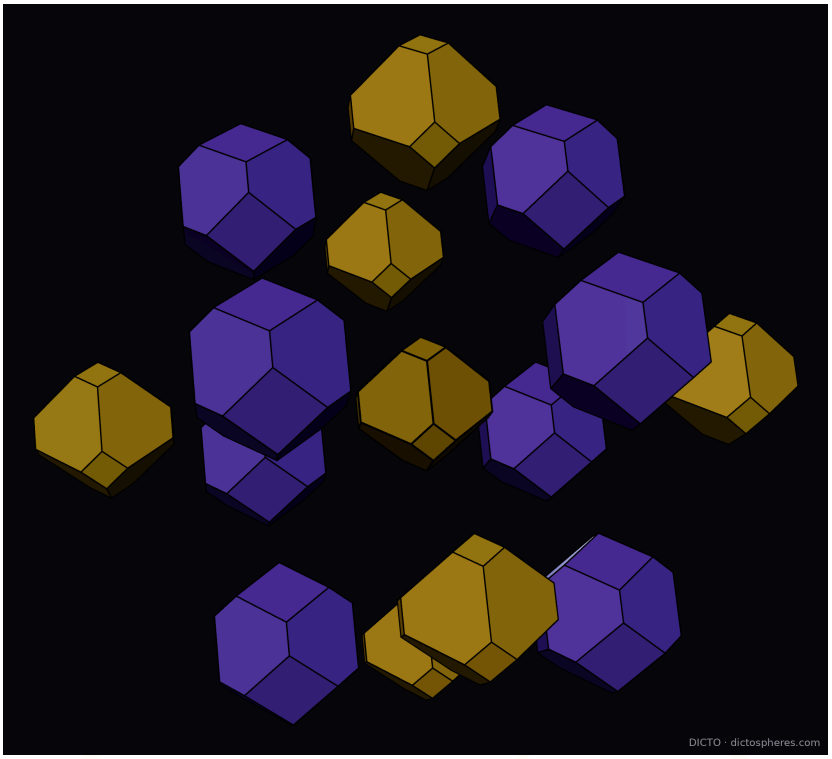


Figure 4. The golden-bond filling at x = 1/2 pulled apart round one DICTO 1:2 Truncated Octahedron (gold, centre): its eight DICTO 2:1 partners (violet) and its six nearest 1:2 neighbours, each moved straight out from the centre.

6. Leads

(1) The Vacuus of other simple 4-polytopes, the Queens of #104 first: how many central slices carry their faces, and which of them are 3D solids already in the work. (2) The golden slice of the permutohedron honeycomb (the 4D permutohedra fill 4-space) as a non-repeating 3D filling: its shapes counted exactly. (3) The heart for seven values: the 6D permutohedron holds seven 5D permutohedra of 1, 2, 3, 5, 6, 7; how they are slotted. (4) A physical build of the heart: five card truncated octahedra cannot be slotted in ordinary space, but their ten crossing hexagons can be shown in the app.

Sources checked

G. M. Ziegler, Lectures on Polytopes, Springer GTM 152 (1995), Example 0.10 (the permutohedron, its faces as ordered partitions); R. P. Stanley, On the number of reduced decompositions of elements of Coxeter groups, European J. Combin. 5 (1984) 359–372 (768 = the number of reduced words of 54321); A. Björner and F. Brenti, Combinatorics of Coxeter Groups, Springer GTM 231 (2005) (the weak order; braid and commutation relations); A. Postnikov, Permutohedra, associahedra, and beyond, IMRN 2009 (permutohedra of arbitrary values); J. H. Conway and N. J. A. Sloane, Sphere Packings, Lattices and Groups (the permutohedron as the Voronoi cell of A_n^*); OEIS searches 11 Nov 2026 for 3, 15, 65, 225, 770, 2443 and 12, 50, 210, 707, 2380 (no match), A002426 (central trinomial coefficients); M. Koca, N. O. Koca, M. Al-Mukhaini, A. Al-Qanobi, Quaternionic representations of the pyritohedral group, related polyhedra and lattices, arXiv:1506.04600 (2015), eq. 49 (T(x)); R. Rado, An inequality, J. London Math. Soc. 27 (1952) 1–6, and G. H. Hardy, J. E. Littlewood, G. Pólya, Inequalities (1934) (majorization); arXiv 2509.13877, To cover a permutohedron (covering corners by hyperplanes, a different question); arXiv 1305.6097, Permutonestohedra;

B. A. Venkov, On a class of Euclidean polyhedra, Vestnik Leningrad Univ. 9 (1954), and P. McMullen, Convex bodies which tile space by translation, Mathematika 27 (1980) (a translation tiler has centrally symmetric faces, after H. Minkowski 1897). DICTO's works: #48 (T(x), the golden bond), #104 (the sweep), the Vacuus builds (the octahedron Vacuus in cardboard).